

Residual windows for partial semigroups and graph-directed conservative dynamics

Yitzchak Shmalo

Einstein Institute of Mathematics, The Hebrew University of Jerusalem, Givat Ram, Jerusalem, Israel
yitzchak.shmalo@gmail.com

Abstract

We prove a residual-window transitivity theorem for countable families of partial homeomorphisms. Each generator is given by an open structural homeomorphism, while its admissible domain is only assumed to be comeagre in the structural domain and may have empty interior. If the structural partial semigroup is open-set transitive, then the admissible system is open-set transitive in the branch sense. More precisely, there is a forward-invariant comeagre set of initial points such that, from each of its points, admissible branches can hit any prescribed sequence of nonempty open targets. The dense-open window theorem is recovered as a special case. Combined with the generic transitivity theorem of Koropecski and Nassiri, the result gives residual-window statements for generic conservative surface pairs and finite tuples. A graph-directed formulation gives analogous conclusions, including a private-edge criterion for strongly connected conservative graphs. The hypothesis of comeagreness is sharp: density and full measure do not suffice.

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1 Introduction

The Birkhoff transitivity principle says that open-set transitivity on a suitable Baire phase space produces many points with dense forward orbit; see, for instance, [1, 4]. This paper proves the corresponding principle for maps which are only partially available.

Suppose that $g_1, \dots, g_m$ generate an open-set transitive positive semigroup on a Baire space X , but that g_i may be used only at points of a prescribed set $D_i \subset X$. If the D_i are dense open, then the restrictions $g_i|_{D_i}$ are ordinary partial homeomorphisms between open sets, and one can ask whether they still generate an open-set transitive pseudosemigroup. This dense-open case was treated in [5]. The residual case requires a different formulation: a comeagre set may have empty interior, so the restricted maps need not define a pseudosemigroup with open domains. The point of the present paper is that residual domains nevertheless support a robust admissible-branch form of transitivity.

Our main theorem works in a general partial setting. Each generator has an open structural homeomorphism $\bar{h} : U_h \rightarrow V_h$ and an admissible domain $D_h \subset U_h$ whose complement is meagre in U_h . If the structural words are open-set transitive, then admissible words still send every nonempty open set into every other, a comeagre set of starting points can hit each fixed open target, and a single forward-invariant comeagre set R supports branches hitting any prescribed sequence of open targets in order. The conclusion is therefore uniform over the subsequent choice of target sequence.

For conservative surfaces we use the generic transitivity theorem of Koropecski and Nassiri for positive semigroups generated by area-preserving diffeomorphisms [2, 3]. Combining it with the residual-window theorem gives the following statement: for a residual set of conservative pairs, every subsequent choice of comeagre windows admits residual prescribed-hitting branches. The dense-open results of [5] are retained explicitly as special cases, where the usual restricted pseudosemigroup is again open-set transitive.

The graph-directed and private-edge consequences from the dense-open setting are also included in compact form. A finite directed graph is handled by applying the main theorem to the disjoint union of the vertex spaces. In the conservative private-edge setting, two based loops with private single-use edges allow the two core return maps to be varied independently; applying the Koropecski–Nassiri theorem to the return pair then gives residual prescribed hitting throughout any strongly connected graph, for every subsequent choice of comeagre edge windows.

The examples in Section 5 show that comeagreness is essential: density alone, and even full measure, cannot replace it. Throughout, a subset of a space Y is called *comeagre in Y* if its complement is meagre in Y ; no openness or G_δ assumption is included in this terminology.

2 Residual-domain partial homeomorphisms

We begin with two elementary category facts, included to make clear that no measurability or Borel assumption on the windows is being used.

Lemma 2.1. *Let Y be an open subspace of a topological space X .*

- (i) *If A is meagre in Y , then A is meagre in X .*
- (ii) *If $f : Y \rightarrow Z$ is a homeomorphism onto an open subspace Z of X and L is comeagre in Z , then $f^{-1}(L)$ is comeagre in Y .*

Proof. For (i), it is enough to prove the assertion when A is nowhere dense in Y , because a countable union of sets meagre in X is meagre in X . Let $O \subset X$ be nonempty and open. If $O \cap Y = \emptyset$, then O itself is a nonempty open subset of O disjoint from A , and indeed from $\bar{A}^X$, because every point of the open set O has the neighbourhood O disjoint from $Y \supset A$. If $O \cap Y \neq \emptyset$, then $O \cap Y$ is a nonempty open set in Y . Since A is nowhere dense in Y , there is a nonempty open set W of Y such that

$$W \subset O \cap Y, \quad W \cap \bar{A}^Y = \emptyset.$$

Because Y is open in X , the set W is open in X . Also $W \cap \bar{A}^X = W \cap \bar{A}^Y = \emptyset$, since $W \subset Y$. Thus every nonempty open subset of X contains a nonempty open subset disjoint from $\bar{A}^X$, so $\bar{A}^X$ has empty interior and A is nowhere dense in X .

For (ii), write $Z \setminus L = M$, where M is meagre in Z . A homeomorphism maps nowhere dense subsets of Z to nowhere dense subsets of Y after taking inverse images, hence $f^{-1}(M)$ is meagre in Y . Therefore $f^{-1}(L) = Y \setminus f^{-1}(M)$ is comeagre in Y . $\square$

Definition 2.2. A *residual-domain partial homeomorphism* of X is a quadruple

$$h = (U_h, V_h, \bar{h}, D_h),$$

where $U_h, V_h \subset X$ are open, $\bar{h} : U_h \rightarrow V_h$ is a homeomorphism and $D_h \subset U_h$ is comeagre in U_h . We call $\bar{h}$ the structural map and D_h the admissible domain. The actually usable map is the restriction of $\bar{h}$ to D_h .

Let $\mathcal{H}$ be a countable family of residual-domain partial homeomorphisms of X . If

$$w = h_n \cdots h_1$$

is a nonempty word in letters from $\mathcal{H}$, set

$$\Phi_0 = \text{id}, \quad \Phi_j = \bar{h}_j \circ \cdots \circ \bar{h}_1 \quad (1 \leq j \leq n).$$

The structural domain of w is

$$U_w = \{x \in U_{h_1} : \Phi_j(x) \in U_{h_{j+1}} \text{ for } 1 \leq j < n\},$$

and the structural map is $\bar{h}_w = \Phi_n : U_w \rightarrow \bar{h}_w(U_w)$. The admissible domain is

$$D_w = \{x \in U_w : \Phi_{j-1}(x) \in D_{h_j} \text{ for } 1 \leq j \leq n\}.$$

Thus w is admissible at x exactly when $x \in D_w$.

Lemma 2.3. *For every nonempty word w in $\mathcal{H}$, the structural domain U_w is open in X , the structural map $\bar{h}_w : U_w \rightarrow \bar{h}_w(U_w)$ is a homeomorphism onto an open subset of X , and the admissible domain D_w is comeagre in U_w .*

Proof. We prove the structural assertion by induction on the length n of w . For $n = 1$ it is part of the definition. Suppose the assertion is known for $h_{n-1} \cdots h_1$. If $v = h_{n-1} \cdots h_1$, then

$$U_{h_n v} = U_v \cap \bar{h}_v^{-1}(U_{h_n}).$$

The set U_v is open by the induction hypothesis, U_{h_n} is open, and $\bar{h}_v$ is continuous on U_v , so $U_{h_n v}$ is open. The restriction of the homeomorphism $\bar{h}_n \circ \bar{h}_v$ to this open set is a homeomorphism onto

$$\bar{h}_n(\bar{h}_v(U_v) \cap U_{h_n}),$$

which is open in X because $\bar{h}_v(U_v)$ is open in X , U_{h_n} is open in X , and $\bar{h}_n$ is a homeomorphism from U_{h_n} onto the open set V_{h_n} .

It remains to prove the category assertion. For $1 \leq j \leq n$, the map Φ_{j-1} is a homeomorphism from an open structural prefix domain onto an open subset of X . Its restriction to U_w is therefore a homeomorphism from the open set U_w onto the open set $\Phi_{j-1}(U_w) \subset U_{h_j}$. Since D_{h_j} is comeagre in U_{h_j} , the intersection $D_{h_j} \cap \Phi_{j-1}(U_w)$ is comeagre in $\Phi_{j-1}(U_w)$. Lemma 2.1(ii) gives that

$$\Phi_{j-1}^{-1}(D_{h_j}) \cap U_w$$

is comeagre in U_w . Finally

$$D_w = \bigcap_{j=1}^n (\Phi_{j-1}^{-1}(D_{h_j}) \cap U_w),$$

and a finite intersection of comeagre subsets of U_w is comeagre in U_w . $\square$

Definition 2.4. The family $\mathcal{H}$ is *structurally open-set transitive* if, for all nonempty open sets $A, B \subset X$, there is a nonempty word w in $\mathcal{H}$ such that

$$A \cap U_w \cap \bar{h}_w^{-1}(B) \neq \emptyset.$$

It is *admissibly open-set transitive* if, for all nonempty open sets $A, B \subset X$, there is a nonempty word w in $\mathcal{H}$ such that

$$A \cap D_w \cap \bar{h}_w^{-1}(B) \neq \emptyset.$$

Definition 2.5. A branch from $x \in X$ is an infinite sequence $a_1, a_2, \dots$ in $\mathcal{H}$. It is *admissible* if $x \in D_{a_n \dots a_1}$ for every $n \geq 1$, and it is *dense* if the set of prefix images

$$\{x\} \cup \{\bar{a}_n \circ \dots \circ \bar{a}_1(x) : n \geq 1\}$$

is dense in X .

Theorem 2.6 (Residual-window transitivity). *Let X be a second-countable Baire space, and let $\mathcal{H}$ be a countable family of residual-domain partial homeomorphisms of X . If $\mathcal{H}$ is structurally open-set transitive, then the following assertions hold.*

(i) *For every nonempty open set $B \subset X$, the set*

$$Q(B) = \bigcup_w (D_w \cap \bar{h}_w^{-1}(B)),$$

where w ranges over all nonempty words in $\mathcal{H}$, is comeagre in X .

(ii) *The family $\mathcal{H}$ is admissibly open-set transitive.*

(iii) *There is a comeagre set $R \subset X$ which is forward-invariant under all admissible words: if w is a nonempty word and $x \in R \cap D_w$, then $\bar{h}_w(x) \in R$.*

(iv) *For every $x_0 \in R$ and every sequence $G_1, G_2, \dots$ of nonempty open subsets of X , there is one infinite admissible branch $a_1, a_2, \dots$ from x_0 and integers*

$$0 < N_1 < N_2 < \dots$$

such that

$$\bar{a}_{N_j} \circ \dots \circ \bar{a}_1(x_0) \in G_j \quad (j \geq 1).$$

Consequently every point of R admits a dense admissible branch.

Proof. Let $\mathcal{W}$ be the countable set of nonempty words in the alphabet $\mathcal{H}$, and let $(V_i)_{i \geq 1}$ be a countable basis of nonempty open sets for X . For a basis element V_i define

$$Q_i = \bigcup_{w \in \mathcal{W}} (D_w \cap \bar{h}_w^{-1}(V_i)).$$

We first show that Q_i is comeagre in X for every i .

Fix i , and let $(B_j)_{j \geq 1}$ be another enumeration of a countable basis of nonempty open sets. By structural open-set transitivity, for each pair (j, i) there is a word $u(j, i) \in \mathcal{W}$ such that

$$O_{j,i} := B_j \cap U_{u(j,i)} \cap \bar{h}_{u(j,i)}^{-1}(V_i)$$

is nonempty. It is open because $U_{u(j,i)}$ is open and $\bar{h}_{u(j,i)}$ is continuous on its structural domain. The union

$$O_i := \bigcup_{j \geq 1} O_{j,i}$$

is dense and open in X : every nonempty open subset of X contains some basis element B_j , and that B_j contains the nonempty open set $O_{j,i}$.

For each j , Lemma 2.3 says that $D_{u(j,i)}$ is comeagre in $U_{u(j,i)}$. Hence $O_{j,i} \cap D_{u(j,i)}$ is comeagre in $O_{j,i}$. Also

$$O_{j,i} \cap D_{u(j,i)} \subset Q_i,$$

because points in this set are admissible for the word $u(j,i)$ and are sent into V_i . Therefore $O_{j,i} \setminus Q_i$ is meagre in $O_{j,i}$, and by Lemma 2.1(i) it is meagre in X . Since

$$O_i \setminus Q_i \subset \bigcup_{j \geq 1} (O_{j,i} \setminus Q_i),$$

the set $O_i \setminus Q_i$ is meagre in X . The complement $X \setminus O_i$ is nowhere dense because O_i is dense open. Thus

$$X \setminus Q_i \subset (X \setminus O_i) \cup (O_i \setminus Q_i)$$

is meagre, proving that Q_i is comeagre.

Now let $B \subset X$ be an arbitrary nonempty open set. Choose a basis element $V_i \subset B$. Since $Q_i \subset Q(B)$ and Q_i is comeagre, $Q(B)$ is comeagre. This proves (i). Assertion (ii) follows at once: if A, B are nonempty open sets, then $A \cap Q(B)$ is nonempty because A is a Baire space and $Q(B)$ is comeagre in X . Any point of $A \cap Q(B)$ is admissibly sent into B by some word.

It remains to construct the invariant residual set. Put

$$R_0 = \bigcap_{i \geq 1} Q_i.$$

This set is comeagre by the preceding paragraph. For a comeagre set $E \subset X$ and a word $w \in \mathcal{W}$, define

$$P_w(E) = (X \setminus D_w) \cup (D_w \cap \bar{h}_w^{-1}(E)).$$

We claim that $P_w(E)$ is comeagre in X . Let $M = X \setminus E$, which is meagre. Since $\bar{h}_w : U_w \rightarrow \bar{h}_w(U_w)$ is a homeomorphism onto an open subspace and $M \cap \bar{h}_w(U_w)$ is meagre in that open subspace, Lemma 2.1(ii) implies that

$$\bar{h}_w^{-1}(M \cap \bar{h}_w(U_w))$$

is meagre in U_w , and hence meagre in X by Lemma 2.1(i). But

$$X \setminus P_w(E) = D_w \cap \bar{h}_w^{-1}(M \cap \bar{h}_w(U_w))$$

is contained in that meagre set. Thus $P_w(E)$ is comeagre.

Define recursively

$$R_{k+1} = R_k \cap \bigcap_{w \in \mathcal{W}} P_w(R_k) \quad (k \geq 0),$$

and put

$$R = \bigcap_{k \geq 0} R_k.$$

Every R_k is comeagre, since $\mathcal{W}$ is countable and each $P_w(R_k)$ is comeagre. Hence R is comeagre. If $x \in R \cap D_w$, then for every k we have $x \in P_w(R_k)$. The definition of $P_w(R_k)$ and the assumption $x \in D_w$ give $\bar{h}_w(x) \in R_k$ for all k . Therefore $\bar{h}_w(x) \in R$, proving (iii).

Finally fix $x_0 \in R$ and nonempty open targets $G_1, G_2, \dots$. Choose basis elements $V_{i_j} \subset G_j$. We construct finite words w_j and points x_j inductively. Suppose $x_{j-1} \in R$ has been constructed. Because $R \subset R_0 \subset Q_{i_j}$, there is a word $w_j \in \mathcal{W}$ such that

$$x_{j-1} \in D_{w_j}, \quad x_j := \bar{h}_{w_j}(x_{j-1}) \in V_{i_j} \subset G_j.$$

By the forward-invariance already proved, $x_j \in R$. This completes the induction.

Write each word w_j as a finite product of letters from $\mathcal{H}$, and concatenate these finite blocks. The resulting infinite sequence of letters is an admissible branch from x_0 , because every block is admissible at the point reached before that block. If N_j is the total length of the first j blocks, then the N_j th prefix image is $x_j \in G_j$. This proves (iv). Taking the targets G_j to enumerate the basis gives a dense admissible branch. $\square$

Remark 2.7. Theorem 2.6 is the stronger transitivity statement that remains valid for arbitrary comeagre windows. It gives admissible open-set transitivity, residual hitting into every open target, and prescribed hitting from every point of one invariant residual set. What cannot generally be strengthened, without assuming the windows are open, is ordinary open-set transitivity of a pseudosemigroup between open domains: a comeagre window may have empty interior, so no nonempty open domain for the restricted map need exist.

3 Global semigroups and conservative surfaces

We first spell out the consequence for globally defined generators.

Corollary 3.1. *Let X be a second-countable Baire space, and let $g_1, \dots, g_m \in \text{Homeo}(X)$ generate an open-set transitive positive semigroup. Let $D_1, \dots, D_m \subset X$ be comeagre. Then the following hold.*

- (i) *For every pair of nonempty open sets $A, B \subset X$, there is a positive word $w = g_{i_n} \cdots g_{i_1}$ and a point $x \in A$ such that*

$$x \in D_w, \quad w(x) \in B,$$

where $w_0 = \text{id}$, $w_k = g_{i_k} \cdots g_{i_1}$ for $1 \leq k \leq n$, and

$$D_w = \bigcap_{k=0}^{n-1} w_k^{-1}(D_{i_{k+1}}).$$

- (ii) *There is a comeagre set $R_D \subset X$ such that for every $x \in R_D$ and every sequence of nonempty open targets $G_1, G_2, \dots$, there are generator indices $i_1, i_2, \dots$ and integers $0 < N_1 < N_2 < \dots$ such that, with*

$$z_0 = x, \quad z_n = g_{i_n}(z_{n-1}) \quad (n \geq 1),$$

one has

$$z_{n-1} \in D_{i_n} \quad (n \geq 1), \quad z_{N_j} \in G_j \quad (j \geq 1).$$

In particular, every $x \in R_D$ admits a dense admissible branch.

(iii) If the windows D_i are dense open, then the ordinary partial homeomorphisms $g_i|_{D_i} : D_i \rightarrow g_i(D_i)$ generate an open-set transitive pseudosemigroup.

Proof. Regard each g_i as a residual-domain partial homeomorphism with structural domain and structural image equal to X , structural map g_i , and admissible domain D_i . The structural words are exactly the positive semigroup words in the g_i . Hence Theorem 2.6 gives (i) and (ii).

Assume now that each D_i is dense open. Given nonempty open $A, B \subset X$, choose a positive word $w = g_{i_n} \cdots g_{i_1}$ with

$$A \cap w^{-1}(B) \neq \emptyset.$$

Set $w_0 = \text{id}$ and $w_k = g_{i_k} \cdots g_{i_1}$ for $1 \leq k \leq n$. The set $A \cap w^{-1}(B)$ is nonempty and open. For each k , the set $w_k^{-1}(D_{i_{k+1}})$, interpreted for $0 \leq k < n$, is dense open in the appropriate open domain because w_k is a homeomorphism. Therefore

$$A \cap w^{-1}(B) \cap \bigcap_{k=0}^{n-1} w_k^{-1}(D_{i_{k+1}})$$

is a finite intersection of a nonempty open set with dense open subsets of that open set. Since open subspaces of a Baire space are Baire, this intersection is nonempty. Any point in it follows the restricted partial maps from A into B , proving open-set transitivity of the restricted pseudosemigroup. $\square$

Let (S, ω) be a closed connected orientable surface with area form, and let $\text{Diff}_\omega^r(S)$ denote the group of C^r area-preserving diffeomorphisms, where $r \in \mathbb{N} \cup \{\infty\}$ and $r \geq 1$ when r is finite.

Corollary 3.2. *For every $r \in \mathbb{N} \cup \{\infty\}$ there is a dense G_δ subset*

$$\mathcal{R}_2^r(S, \omega) \subset \text{Diff}_\omega^r(S) \times \text{Diff}_\omega^r(S)$$

with the following property. If $(g_0, g_1) \in \mathcal{R}_2^r(S, \omega)$ and $D_0, D_1 \subset S$ are comeagre, then the windowed system is admissibly open-set transitive and there is a comeagre set $R_D \subset S$ such that every $x \in R_D$ admits admissible branches hitting any prescribed sequence of nonempty open targets in order. In particular, every $x \in R_D$ admits a dense admissible branch generated by g_0, g_1 and respecting the windows D_0, D_1 .

If D_0, D_1 are dense open, then $g_0|_{D_0}$ and $g_1|_{D_1}$ generate an open-set transitive pseudosemigroup.

Proof. Let $\mathcal{T}^r$ be the set of pairs in $\text{Diff}_\omega^r(S)^2$ whose positive semigroup is open-set transitive. We first note that $\mathcal{T}^r$ is a G_δ set. Let $(B_i)_{i \geq 1}$ be a countable basis for S , and let $\mathcal{W}_2$ be the countable set of nonempty words in two letters. Then

$$\mathcal{T}^r = \bigcap_{i, j \geq 1} \bigcup_{w \in \mathcal{W}_2} \{(f_0, f_1) : w(f_0, f_1)(B_i) \cap B_j \neq \emptyset\}.$$

For a fixed word w and fixed basis elements B_i, B_j , the set inside braces is open in the C^r topology: if one point of B_i is sent into the open set B_j , then the same remains true for all sufficiently small C^0 , hence sufficiently small C^r , perturbations of the finitely many maps appearing in w . Therefore $\mathcal{T}^r$ is G_δ .

The theorem of Koropecki and Nassiri cited in [2, 3] states that, for every $r \in \mathbb{N} \cup \{\infty\}$, a residual set of pairs of area-preserving surface diffeomorphisms has transitive positive semigroup action. In the compact full-support conservative setting, the equivalent formulations recorded in [2, Lemmas 3–4] give the open-set form used here. Thus $\mathcal{T}^r$ is dense as well as G_δ . Set $\mathcal{R}_2^r(S, \omega) = \mathcal{T}^r$ and apply Corollary 3.1. $\square$

Corollary 3.3 (Finite reservoirs). *Let $m \geq 2$. For every $r \in \mathbb{N} \cup \{\infty\}$ there is a dense G_δ subset*

$$\mathcal{R}_m^r(S, \omega) \subset (\text{Diff}_\omega^r(S))^m$$

such that, for every $(g_1, \dots, g_m) \in \mathcal{R}_m^r(S, \omega)$ and every choice of comeagre windows $D_1, \dots, D_m \subset S$, the windowed m -generator system is admissibly open-set transitive and a comeagre set of starting points admits prescribed-hitting admissible branches. If the windows are dense open, then the restricted m -generator pseudosemigroup is open-set transitive.

Proof. For each pair $1 \leq i < j \leq m$, let π_{ij} be the coordinate projection from $\text{Diff}_\omega^r(S)^m$ to the (i, j) coordinates, and define

$$\mathcal{R}_m^r(S, \omega) = \bigcap_{1 \leq i < j \leq m} \pi_{ij}^{-1}(\mathcal{R}_2^r(S, \omega)).$$

This is a finite intersection of dense G_δ sets, hence is dense G_δ . If $(g_1, \dots, g_m)$ belongs to it, then every coordinate pair generates an open-set transitive positive semigroup. In particular, the whole m -generator positive semigroup is open-set transitive. Corollary 3.1 applies. $\square$

4 Graph-directed residual windows

Let $G = (V, E)$ be a finite directed graph, with source and target maps s, t . For each $v \in V$ let X_v be a second-countable Baire space and put $X = \bigsqcup_{v \in V} X_v$. Since V is finite, X is again second-countable and Baire. A graph-directed residual-domain system assigns to each edge e a residual-domain partial homeomorphism

$$h_e = (U_e, V_e, \bar{h}_e, D_e), \quad U_e \subset X_{s(e)}, \quad V_e \subset X_{t(e)}.$$

A path $p = e_n \cdots e_1$ has the structural and admissible data defined by composition as in Section 2; empty paths, used only for transport, mean the identity on the relevant component. The system is structurally open-set transitive if, for every nonempty open $A \subset X_u$ and $B \subset X_v$, some path $p : u \rightarrow v$ satisfies

$$A \cap U_p \cap \bar{h}_p^{-1}(B) \neq \emptyset.$$

Corollary 4.1 (Graph-directed windows and private-edge forcing). *The following hold.*

- (i) *If a finite graph-directed residual-domain system is structurally open-set transitive, then it is admissibly open-set transitive. Moreover, for each vertex v there is a comeagre set $R_v \subset X_v$ such that every $x \in R_v$ admits admissible branches hitting any prescribed sequence of nonempty open targets in prescribed components. If all edge windows D_e are dense open in their structural domains, then the ordinary restricted edge maps generate an open-set transitive graph-directed pseudosemigroup.*
- (ii) *Fix $r \in \mathbb{N} \cup \{\infty\}$, with $r \geq 1$ when r is finite. Let (S, ω) be a closed connected orientable surface, let $H = \text{Diff}_\omega^r(S)$, and let G be finite and strongly connected, with every vertex carrying a copy of S and every edge carrying a global map in H . Suppose two based loop words γ_0, γ_1 at a vertex v_* have private single-use edges $\varepsilon_0, \varepsilon_1$: the edge ε_i occurs exactly once in γ_i and not in γ_{1-i} . If all non-private edge maps are fixed, then a dense G_δ set of choices of $(h_{\varepsilon_0}, h_{\varepsilon_1}) \in H^2$ has the following universal property: for every assignment of comeagre edge windows, the resulting graph-directed system is admissibly open-set transitive and has the prescribed-hitting residual sets from (i). The same property holds for a residual subset of the full edge-parameter space H^E .*

Proof. For (i), regard each edge map as a residual-domain partial homeomorphism of the disjoint union X . The edge family is finite, and graph-directed structural transitivity is exactly structural open-set transitivity on X . Theorem 2.6 gives a forward-invariant comeagre set $R \subset X$ with prescribed hitting; set $R_v = R \cap X_v$, which is comeagre in X_v . If the D_e are dense open, then along any structural path $p = e_n \cdots e_1$ the admissible domain

$$D_p = U_p \cap \bigcap_{k=0}^{n-1} \Phi_k^{-1}(D_{e_{k+1}})$$

is dense open in U_p , where Φ_k is the k th structural prefix. Intersecting D_p with a nonempty open structural hitting set gives the dense-open pseudosemigroup conclusion.

For (ii), fix the non-private maps. Because ε_i appears once in γ_i and not in the other loop, the return map along γ_i has the form

$$R_i = A_i \circ h_{\varepsilon_i} \circ B_i \quad (i = 0, 1)$$

with fixed $A_i, B_i \in H$. Hence $(h_{\varepsilon_0}, h_{\varepsilon_1}) \mapsto (R_0, R_1)$ is a homeomorphism of H^2 . The inverse image of the dense G_δ set $\mathcal{R}_2^r(S, \omega)$ from Corollary 3.2 is therefore dense G_δ . For such a choice, the two structural return maps at v_* are open-set transitive. Strong connectivity and global edge maps give full-domain transport paths from any vertex to v_* and from v_* to any vertex; composing transport in, one return loop, and transport out proves structural open-set transitivity of the graph system. Part (i) then applies to every later choice of comeagre windows.

For the full parameter statement, the good set is G_δ because the return-pair map $H^E \rightarrow H^2$ is continuous. It is dense by the preceding argument on every product basic open set: first choose the non-private coordinates inside the prescribed open sets, then choose the two private coordinates from the corresponding dense section. $\square$

Remark 4.2. Part (i) contains the deterministic graph-directed dense-open propagation theorem of [5], and part (ii) contains its conservative private-edge forcing theorem, now strengthened from dense-open windows to arbitrary comeagre windows. The private-edge condition is weaker than edge-disjointness of the two loops; the loops may overlap away from the two private single-use edges. The conclusion is universal in the windows: the edge maps are chosen before the comeagre windows and before the target sequence are specified.

5 Sharpness of the window hypothesis

The first example shows that topological density is too weak.

Proposition 5.1. *The comeagreness of the windows in Theorem 2.6 and Corollary 3.1 cannot be replaced by density.*

Proof. Let $X = \mathbb{R}/\mathbb{Z}$ and let $g(x) = x + \alpha$, where $\alpha \notin \mathbb{Q}/\mathbb{Z}$. The positive semigroup generated by g is open-set transitive. Let $D = \mathbb{Q}/\mathbb{Z}$. This set is dense. However, if $x \in D$ and $g(x) \in D$, then

$$\alpha = g(x) - x \in \mathbb{Q}/\mathbb{Z},$$

a contradiction. Thus $D \cap g^{-1}(D) = \emptyset$. Consequently the restriction of g to the dense window D has no admissible branch of length two, and in particular no dense infinite admissible branch. $\square$

The second example separates residual largeness from measure-theoretic largeness.

Proposition 5.2. *Let λ be Haar probability measure on $\mathbb{T}^2$. There are smooth Haar-measure-preserving diffeomorphisms g_0, g_1 of $\mathbb{T}^2$ and full-measure meagre sets $D_0, D_1 \subset \mathbb{T}^2$ such that g_0, g_1 generate a positive semigroup all of whose unwindowed orbits are dense, but the set of initial points admitting a dense admissible branch respecting D_0, D_1 is meagre. Nevertheless, for every nonempty open $U \subset \mathbb{T}^2$, almost every point has some admissible positive word reaching U .*

Proof. Choose irrational numbers α, β and set

$$g_0(x, y) = (x + \alpha, y), \quad g_1(x, y) = (x, y + \beta).$$

For every $(x, y) \in \mathbb{T}^2$, the positive orbit

$$\{(x + n\alpha, y + m\beta) : n, m \in \mathbb{N} \cup \{0\}\}$$

is dense in $\mathbb{T}^2$, because the one-dimensional positive rotation orbits of α and β are dense in $\mathbb{T}$.

We construct a full-measure meagre set D . Let $q_1, q_2, \dots$ be a countable dense subset of $\mathbb{T}^2$. For each n , choose a dense open set O_n with

$$\lambda(O_n) < 2^{-n} \quad \text{and} \quad O_n \cap \{q_1, \dots, q_n\} = \emptyset.$$

This is possible by taking a union of sufficiently small open balls around a countable dense subset of $\mathbb{T}^2 \setminus \{q_1, \dots, q_n\}$. Put

$$K_n = \mathbb{T}^2 \setminus O_n, \quad D = \bigcup_{n \geq 1} K_n.$$

Each K_n is closed and has empty interior, because O_n is dense. Hence D is meagre. On the other hand,

$$\lambda(D) \geq \lambda(K_n) > 1 - 2^{-n}$$

for every n , so $\lambda(D) = 1$.

Set $D_0 = D_1 = D$. Any nonempty admissible branch must start in D , so the set of points admitting a dense admissible branch is contained in the meagre set D .

Finally fix a nonempty open set $U \subset \mathbb{T}^2$. For a positive word $w = g_{i_N} \cdots g_{i_1}$, its admissibility domain is

$$A_w = \bigcap_{k=0}^{N-1} w_k^{-1}(D_{i_{k+1}}), \quad w_0 = \text{id},$$

where w_k is the k th prefix of w . Each A_w is conull because the translations preserve λ and D_0, D_1 are conull. Since every unwindowed positive orbit is dense, for every point $x \in \mathbb{T}^2$ there is at least one positive word w with $w(x) \in U$; equivalently,

$$\mathbb{T}^2 = \bigcup_w w^{-1}(U),$$

where the union is over the countable set of positive words. The set of points for which no admissible word reaches U is therefore contained in

$$\bigcup_w (w^{-1}(U) \setminus A_w),$$

a countable union of null sets. Hence almost every point reaches U by some admissible positive word. $\square$

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Declaration of use of AI-assisted tools

During the preparation of this manuscript, the author used generative AI tools to assist with drafting and revision. All mathematical claims, proofs and bibliographic information were subsequently reviewed and edited by the author, who takes full responsibility for the content of the manuscript.

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